Similar questions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 It is given that θ satisfies $\sin\left(2\theta + \frac{1}{6}\pi\right) = 2\cos\left(2\theta + \frac{1}{6}\pi\right)$.

(a) Show that $\tan 2\theta = 5\sqrt{3} - 8$. **[5]**

(b) Hence find the values of θ in the interval $0 < \theta < \pi$, giving your answers in radians to 3 decimal places. **[3]**

2 The size P of a population of birds, t months after the start of a study, is modelled by $P = 2000 + 500\sin(0.5t)$ for $t \geqslant 0$, where t is in radians.

(a) Write down the maximum and minimum populations predicted by the model, and the time between successive maxima. **[3]**

(b) Find the times in the first cycle for which $P \geqslant 2250$, and the length of time for which this is true. **[4]**

(c) Observations show that the fluctuations get smaller over time. Suggest a modification to the model to include this. **[1]**

Question 1

(a) Dividing by $\cos\left(2\theta+\frac{\pi}{6}\right)$: $\tan\left(2\theta+\frac{\pi}{6}\right)=2$. $\dfrac{\tan 2\theta+\tan\frac{\pi}{6}}{1-\tan 2\theta\tan\frac{\pi}{6}}=2$ with $\tan\frac{\pi}{6}=\dfrac{1}{\sqrt{3}}$: $\tan 2\theta+\dfrac{1}{\sqrt{3}}=2-\dfrac{2\tan 2\theta}{\sqrt{3}}$, so $\tan 2\theta\left(1+\dfrac{2}{\sqrt{3}}\right)=2-\dfrac{1}{\sqrt{3}}$ and $\tan 2\theta=\dfrac{2\sqrt{3}-1}{\sqrt{3}+2}=\dfrac{(2\sqrt{3}-1)(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})}=5\sqrt{3}-8$

(b) $2\theta=\tan^{-1}(5\sqrt{3}-8)=0.585$ or $0.585+\pi$ (in range $0<2\theta<2\pi$). So $\theta=0.292$ or $\theta=1.863$

Question 2

(a) Maximum 2500, minimum 1500. The period is $\dfrac{2\pi}{0.5}=4\pi=12.6$ months

(b) $500\sin(0.5t)\geqslant 250$, so $\sin(0.5t)\geqslant\frac{1}{2}$, i.e. $\frac{\pi}{6}\leqslant 0.5t\leqslant\frac{5\pi}{6}$, so $\frac{\pi}{3}\leqslant t\leqslant\frac{5\pi}{3}$: from $t=1.05$ to $t=5.24$ months, a duration of $\frac{4\pi}{3}=4.19$ months

(c) Multiply the oscillating term by a decaying exponential, e.g. $P=2000+500\mathrm{e}^{-kt}\sin(0.5t)$